Business Requirements Document

IT Metrics Dashboard Project

TechServe Solutions

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EXECUTIVE SUMMARY

Overview

TechServe Solutions, a mid-sized IT services organization with 250 employees, currently lacks real-time visibility into IT service desk operations. The IT Metrics Dashboard is a web-based analytics platform that will aggregate incident data from ServiceNow, calculate critical performance indicators, and provide real-time visibility to IT leadership and operations teams.

Business Problem

Currently, the IT Service Desk at TechServe Solutions operates with significant operational blind spots:

Current Situation:

- Incident data exists in ServiceNow but is not actionable in real-time
- Leadership relies on manual weekly reports compiled in Excel (6-8 hours of effort)
- Reports lag by 1-3 days after the incident period ends
- No visibility into ongoing incident status or SLA compliance until reports are generated
- Help desk manager cannot quickly identify bottlenecks or escalate SLA-at-risk incidents
- IT Director lacks data-driven insights for resource planning and operational improvements

Impact:

- **MTTR (Mean Time to Resolution):** Currently averaging 10.2 hours; incidents are taking longer to resolve due to lack of real-time visibility and coordination
- **SLA Compliance:** Averaging 82%, below the 95% target; violations discovered days later in reports
- **Team Performance Visibility:** Unknown; manager relies on anecdotal feedback
- **Inefficient Resource Use:** No data showing which teams are overloaded or underutilized
- **Slow Decision-Making:** Leaders make operational decisions with week-old data

Proposed Solution

Build an interactive, real-time dashboard that:

1. **Automatically pulls incident data from ServiceNow** every 30 minutes
2. **Calculates critical metrics in real-time:** MTTR, SLA compliance %, incident volume, team performance
3. **Displays metrics through interactive visualizations:** KPI cards, trend charts, performance tables
4. **Enables instant analysis:** Filter by date, priority, category, team
5. **Supports stakeholder communication:** One-click export to Excel/PDF
6. **Alerts on SLA violations:** Notifies manager when incidents approach SLA deadline

Expected Business Benefits

Benefit	**Baseline**	**Target**	**Impact**
Mean Time to Resolution (MTTR)	10.2 hours	7.0 hours	31% improvement; faster incident resolution
SLA Compliance	82%	95%+	Better customer/stakeholder satisfaction
Reporting Time	6-8 hours/week	10 minutes/week	50+ hours/month saved for help desk
Leadership Insight Latency	1-3 days	Real-time	Immediate visibility for proactive response
Team Performance Visibility	Subjective	Data-driven	Better resource allocation decisions
Incident Escalation Response	Hours (via report)	Minutes (real-time alert)	Faster SLA compliance response

Financial Impact

Investment Required:

- Development: 300 hours @ $100/hour = $30,000
- Infrastructure/Hosting: $500/month = $6,000/year
- Maintenance Year 1: 100 hours @ $100/hour = $10,000

  Total Year 1 Cost: $46,000

Annual Benefit:

- Help desk reporting time savings: 300 hours × $50/hour = $15,000
- MTTR improvement impact: Reduced customer downtime = $80,000+
- Improved resource allocation: Better utilization saves = $40,000+

Total Annual Benefit: $135,000+

ROI Analysis:

- Payback Period: 4-5 months
- Year 1 Net Benefit: $89,000
- 3-Year Total Benefit: $355,000

Timeline & Phasing

- **Week 1-2:** MVP (minimum viable product) with core metrics
- **Week 3-4:** Polish, testing, ServiceNow integration
- **Week 5-8:** Full rollout, training, optimization
- **Launch Target:** End of April 2024

Success Criteria

- MTTR reduced to ≤7 hours within 6 months
- SLA compliance maintained at 95%+
- 80%+ adoption by IT staff within 1 month
- Help desk reporting time reduced by 50%
- Dashboard uptime of 99.5%

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1. BUSINESS CONTEXT

1.1 Organization Profile

Company: TechServe Solutions
Organization Type: Mid-sized IT Services Company
Total Employees: 250
IT Department Size: 18 people

- Service Desk Manager: 1
- Help Desk Agents: 12
- Technical Specialists: 4
- IT Director: 1

Industry: IT Services / Managed Services
Location: Multiple office locations (headquarters + 2 remote centres)
Primary Service: Managed IT services for small-to-medium business clients

1.2 Current Environment

Current Tools & Systems:

- **ServiceNow:** Primary incident management system (contains all incident data)
- **Excel Spreadsheets:** Manual metric compilation and reporting
- **Email:** Status communication to stakeholders

- **Confluence:** Internal documentation
- **Active Directory:** User and team management

Current Process for Incident Reporting:

Day 1-7: Incidents happen and are logged in ServiceNow ↓ Friday EOD: Help desk manager exports incident data from ServiceNow ↓ Friday Night-Saturday: Manager manually compiles Excel report

- Calculates MTTR (sum of resolution hours / count)
- Calculates SLA compliance (count of compliant / total)
- Creates charts
- Writes analysis comments ↓ Monday Morning: Report emailed to IT Director and stakeholders ↓ Result: Leadership has 3–5 days old data for decision-making

Current Incident Handling (Baseline Data from Sample):

From our sample data of 200 incidents (Jan-Apr 2024):

- **Total Incidents:** 200
- **By Priority:** Critical: 25 | High: 50 | Medium: 75 | Low: 50
- **By Category:** Network: 50 | Access: 40 | Hardware: 60 | Software: 50
- **Current MTTR:** 10.2 hours average
    - Critical: 1.8 hours
    - High: 5.1 hours
    - Medium: 8.3 hours
    - Low: 28.5 hours
- **Current SLA Compliance:** 82%
    - Critical (2h SLA): 88% compliant
    - High (8h SLA): 84% compliant
    - Medium (24h SLA): 81% compliant
    - Low (72h SLA): 76% compliant

Current Challenges & Pain Points:

Pain Point	Frequency	Impact	Severity
Manual data entry errors	Daily	Metrics inaccurate	High
Email incidents lost	Weekly	Some issues untracked	Medium
No real-time SLA visibility	Continuous	Late response to violations	Critical
Manual metric calculation	Weekly	6-8 hours/week help desk time	High
Report lag (1-3 days)	Always	Decisions made on old data	Critical

Pain Point	Frequency	Impact	Severity
No drill-down capability	On-demand	Cannot quickly diagnose issues	Medium
Inconsistent categorization	Daily	Metrics unreliable	Medium
Team workload unknown	Always	Reactive staffing decisions	High

1.3 Stakeholders & Decision Makers

IT Director - Michael Rodriguez (Budget owner, Approver)

- Needs: Strategic visibility, ROI metrics, SLA compliance proof
- Pain point: Weekly reports take time; no drill-down capability
- Success metric: Dashboard shows 95%+ SLA compliance trending up

Service Desk Manager - Sarah Chen (Primary user, Day-to-day operator)

- Needs: Real-time metrics, team performance, alert capability
- Pain point: Manual reporting takes 6-8 hours; cannot see ongoing issues
- Success metric: Identify SLA-at-risk incidents in minutes, reduce MTTR to 7 hours

Help Desk Team (Supporting users, Secondary users)

- Needs: Understanding how they're doing, self-service visibility
- Pain point: No feedback on performance; metrics feel abstract
- Success metric: Can see personal/team metrics anytime

Finance Department (Budget approval)

- Needs: ROI justification, clear cost-benefit
- Pain point: Manual effort doesn't appear in budget planning
- Success metric: Clear $89K first-year net benefit

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2. BUSINESS REQUIREMENTS

2.1 Primary Business Objectives

BR 001: Reduce Mean Time to Resolution (MTTR)

Goal: Improve incident resolution speed by 30%
Baseline: Current MTTR = 10.2 hours
Target: MTTR ≤ 7.0 hours within 6 months
Metric: Monthly average MTTR from dashboard

How Dashboard Achieves This:

- Real-time visibility: Managers see incidents exceeding SLA thresholds and can escalate immediately

- Team workload visibility: Manager can re-assign tickets from overloaded technicians to available ones
- Performance trending: Can identify which categories/teams are slow and provide targeted coaching
- Historical analysis: Can identify patterns (e.g., "Network incidents take 2x longer") and address root causes

Expected Impact:

- Faster escalation response saves ~3 hours per incident on average
- Better load balancing prevents bottlenecks
- Targeted training improves efficiency
- Result: 30% reduction in MTTR → $80K+ customer impact

BR 002: Maintain/Improve SLA Compliance

Goal: Achieve and sustain 95%+ SLA compliance
Baseline: Current SLA compliance = 82%
Target: 95%+ within 3 months, sustained
Metric: Dashboard SLA compliance % by priority

How Dashboard Achieves This:

- Real-time alerts: Manager notified when critical/high incidents approach SLA deadline
- SLA tracking by priority: Can see which priority levels are failing and why
- Proactive response: Can escalate or re-assign before SLA violation occurs
- Post-mortem analysis: Can review why SLA violations happened and prevent recurrence

Expected Impact:

- Early warning system prevents SLA violations
- Proactive escalation keeps incidents on track
- Easier to explain SLA performance to stakeholders
- Result: 95%+ compliance for customer satisfaction

BR 003: Eliminate Manual Reporting Effort

Goal: Reduce help desk reporting time by 50%
Baseline: Manual reporting takes 6-8 hours/week
Target: Dashboard review takes ≤30 minutes/week
Metric: Help desk hours spent on reporting

How Dashboard Achieves This:

- Real-time metrics: No need to wait until Friday to compile
- Automated calculations: MTTR, SLA %, counts all calculated automatically
- One-click export: Generate professional reports without manual formatting
- Self-service for stakeholders: Leadership can access anytime without requesting from help desk

Expected Impact:

- 50+ hours/month freed up for actual incident resolution
- Better morale (less administrative work)
- Faster response to ad-hoc reporting requests
- Result: $15K annual labour cost savings

BR 004: Enable Data-Driven Decision Making

Goal: Provide leadership with real-time, actionable operational data
Baseline: Decisions made with week-old data, limited drill-down
Target: Real-time metrics with instant analysis capability
Metric: Leadership adoption rate, decision speed improvement

How Dashboard Achieves This:

- Real-time KPI cards: MTTR, SLA %, incident count visible instantly
- Historical trending: Can see weekly/monthly trends for pattern analysis
- Drill-down analysis: Click on category to see all Network incidents and their MTTR
- Filtering: Analyze specific time periods, teams, or priorities
- Export capability: Take data to strategic planning meetings

Expected Impact:

- Faster response to operational issues (minutes vs. days)
- Better resource planning based on real data
- Proactive improvements vs. reactive firefighting
- Result: Improved operational maturity

2.2 Functional Requirements

FR 001: Display Real-Time KPI Metrics

KPI Card 1: Total Incidents (Last 30 Days)

- Display: Large number showing incident count
- Breakdown: Total / Open / Resolved
- Sparkline: Trend line showing daily count over 30 days
- Target: 150-200 incidents per month (normal range)
- Update frequency: Real-time (updates every 30 seconds)

KPI Card 2: Average MTTR

- Display: Mean Time To Resolution in hours
- Trend: Comparison to previous month (↑ or ↓)
- Benchmark: Show target (7.0 hours) vs. actual

- Status: Color-coded (Green <7h, Yellow 7-8h, Red >8h)
- Filters: Respects date range filter
- Example: "7.2 hours (↓ from 7.8h last month) - 3% improvement"

KPI Card 3: SLA Compliance %

- Display: Percentage with status color
  - Green: ≥95% (exceeding target)
  - Yellow: 85-95% (at risk)
  - Red: <85% (failing)
- Trend: % point improvement/decline from previous month
- Breakdown available: SLA % by priority (Critical/High/Medium/Low)
- Example: "92.3% (↑ from 88.5% last month)"

KPI Card 4: Team Efficiency Score

- Display: Custom metric combining MTTR + SLA + volume
- Scale: 0-100 (100 = perfect)
- By Team: Show score for each team
- Trend: Improvement over time
- Example: "Network Team: 87/100 (↑ from 82)"

FR 002: Interactive Charts for Visualization

Chart 1: Incidents Over Time (Line Chart)

- X-axis: Date (daily buckets)
- Y-axis: Incident count
- Lines: Total / Open / Resolved
- Time ranges: Last 7/30/90 days, custom date range
- Interactivity: Click on date to drill down to incident list for that day
- Use case: See if incident volume trending up (capacity planning)

Chart 2: Incidents by Priority (Bar Chart)

- X-axis: Priority level (Critical, High, Medium, Low)
- Y-axis: Count and percentage
- Color-coded: Red (Critical), Orange (High), Yellow (Medium), Green (Low)
- Data labels: Show both count and %
- Interactivity: Click priority to filter all other charts to that priority
- Use case: Understand workload distribution

Chart 3: Incidents by Category (Pie Chart)

- Slices: Network, Access, Hardware, Software, Other
- Labels: Category name, count, percentage
- Table view option: For accessibility/export
- Interactivity: Click slice to filter; hover for tooltip
- Use case: Where is effort going (which infrastructure area needs attention)

Chart 4: MTTR Trend (Line Chart)

- X-axis: Week
- Y-axis: MTTR hours
- Lines: Overall / Critical / High / Medium / Low
- Trend line: Show improvement/decline
- Target benchmark: Show 7.0-hour target line
- Use case: Track improvement over time; see if interventions are working

Chart 5: Team Performance Comparison (Bar Chart)

- X-axis: Team name
- Y-axis: Average MTTR
- Color: Red if above 7h, green if below
- Data labels: Show actual hours
- Interactivity: Click team to see their incident details
- Use case: Identify underperforming teams that need support

FR 003: Filtering & Drill-Down Analysis

Filter 1: Date Range Selector

- Pre-sets: Last 7 days | Last 30 days | Last 90 days | Custom range
- Calendar picker: For custom selection
- Applies to: ALL metrics and charts
- Persistence: Remember selected range for user session
- Use case: Analyze specific periods (after deployment, end of quarter, etc.)

Filter 2: Priority Filter

- Multi-select: Critical | High | Medium | Low
- Default: All selected
- Applies to: All metrics and charts
- Use case: Focus on critical incidents, or analyze low-priority workload

Filter 3: Category Filter

- Multi-select: Network | Access | Hardware | Software | Other
- Default: All selected
- Applies to: All metrics and charts
- Use case: Analyse Network incidents specifically, etc.

Filter 4: Team Filter

- Dropdown: All teams | Network Team | Access Team | Desktop Team | Applications Team
- Default: All teams
- Applies to: All metrics and charts
- Use case: Manager reviews their team's performance

Filter Behaviour:

- Filters are cumulative (Date AND Priority AND Category AND Team)
- Apply button or auto-apply as selections change
- Clear all button to reset to defaults
- Show count of incidents matching current filters

FR 004: Reporting & Export Capability

Export to Excel:

- Filename: "IT_Metrics_Report_[Date].xlsx"
- Sheets included:
    - Summary: KPI metrics, key findings
    - Detailed Data: All incident records matching filters
    - Charts: Chart images (4-5 charts)
    - Metrics by Priority: Breakdown of MTTR, SLA, count by priority
    - Metrics by Team: Team performance summary
- Formatting: Professional, ready to share
- Include: Current filter selections noted in summary
- Use case: Share with executives, include in board presentations

Export to PDF:

- Filename: "IT_Metrics_Report_[Date].pdf"
- Format: 2-page executive summary
- Content: Title page with date, Key metrics (MTTR, SLA, count), Charts, Key insights
- Professional: Company branding, colored metrics, summary commentary
- Use case: Email to stakeholders, board meetings

Export Button:

- Location: Dashboard header (visible on all pages)
- Respects current filters: Exports only data shown in current view
- Shows status: "Generating..." → "Ready to download"

FR 005: Data Refresh & Freshness

Automatic Data Refresh:

- Frequency: Every 30 minutes
- Trigger: ServiceNow API call for new/updated incidents
- Background: Happens automatically; no user action needed
- Visual feedback: "Last updated: 5 minutes ago" shown at bottom of dashboard

Manual Refresh Button:

- Location: Dashboard header
- Function: Force immediate update from ServiceNow
- Use case: Manager wants latest data NOW for urgent decision

Data Freshness Indicator:

- Show timestamp of last update
- Show time since last update
- Color warning if >1 hour stale (in case of refresh failure)

2.3 Non-Functional Requirements

NFR 1: Dashboard Performance & Load Time

Requirement

The IT Metrics Dashboard shall load completely within 5 seconds under normal operating conditions for all authorized users on standard corporate broadband connectivity.

KPI Card 1: Dashboard Initial Load Time

- Target: ≤ 5 seconds
- Measurement Method: Average page load duration
- Monitoring Frequency: Continuous monitoring
- Threshold:
    - Green: < 3 sec
    - Yellow: 3–5 sec
    - Red: > 5 sec

KPI Card 2: Filter Response Time

- Target: < 2 seconds

- Scope: Date, Priority, Category, and Team filters
- Outcome: Faster operational analysis and improved usability

KPI Card 3: Export Generation Time

- Excel Export: ≤ 10 seconds
- PDF Export: ≤ 15 seconds
- Outcome: Enables quick stakeholder reporting and management review

Business Impact

- Reduces user frustration and operational delays
- Enables real-time decision-making during incident escalations
- Improves adoption among IT operations teams

Expected Outcome

- Improved user satisfaction
- Increased dashboard usage and engagement
- Faster access to operational insights

NFR 002: Scalability & Concurrent User Support

Requirement

The dashboard shall support a minimum of 50 concurrent active users without performance degradation.

KPI Card 1: Concurrent User Capacity

- Target: ≥ 50 simultaneous users
- Measurement: Active sessions supported during peak usage

KPI Card 2: System Response Under Load

- Target: < 5 seconds response time during peak usage
- Scope: Dashboard navigation, filtering, and chart rendering

KPI Card 3: Resource Utilization

- CPU Usage Target: < 80%
- Memory Usage Target: < 75%
- Outcome: Stable and reliable platform performance

Business Impact

- Supports multiple IT teams accessing metrics simultaneously
- Prevents system slowdown during operational incidents
- Ensures uninterrupted executive reporting access

Expected Outcome

- Reliable multi-user experience
- Operational continuity during high incident periods
- Improved stakeholder confidence in the platform

NFR 003: Availability & Reliability

Requirement

The IT Metrics Dashboard shall maintain 99.5% system uptime excluding planned maintenance windows.

KPI Card 1: System Uptime Percentage

- Target: ≥ 99.5%
- Measurement Frequency: Monthly

KPI Card 2: Unplanned Downtime

- Target: < 4 hours/month
- Monitoring: Infrastructure and application monitoring tools

KPI Card 3: Recovery Time Objective (RTO)

- Target: < 30 minutes
- Scope: Recovery from application or server failure

Business Impact

- Ensures uninterrupted visibility into IT operations
- Prevents reporting gaps during critical incidents
- Maintains trust in operational monitoring processes

Expected Outcome

- Continuous dashboard accessibility
- Improved operational resilience
- Reduced business disruption during outages

NFR 004: Security & Role-Based Access Control (RBAC)

Requirement

The dashboard shall implement Role-Based Access Control (RBAC) to ensure users can only access data and functionality relevant to their roles.

KPI Card 1: Authorized Access Compliance

- Target: 100% access restricted by user role
- **Roles:**
    - IT Director

  - Service Desk Manager
  - Help Desk Agent
  - Read-Only Stakeholder

KPI Card 2: Unauthorized Access Attempts

- Target: 0 successful unauthorized access incidents
- Monitoring: Security audit logs

KPI Card 3: Audit Logging Coverage

- Target: 100% logging of login attempts and data exports
- Outcome: Improved accountability and traceability

Business Impact

- Protects sensitive operational and incident data
- Ensures compliance with organizational security policies
- Prevents accidental or unauthorized data exposure

Expected Outcome

- Improved data governance
- Secure stakeholder access management
- Reduced security and compliance risks

NFR 005: Data Freshness & Refresh Latency

Requirement

The dashboard shall refresh incident data from ServiceNow with a maximum latency of 30 minutes.

KPI Card 1: Data Refresh Frequency

- Target: Every 30 minutes
- Source: ServiceNow API integration

KPI Card 2: Data Freshness Indicator Accuracy

- Target: 100% accurate timestamp visibility
- Example: “Last updated 12 minutes ago”

KPI Card 3: Refresh Failure Detection

- Target: Alert generated within 5 minutes of failed refresh
- Outcome: Immediate awareness of stale data conditions

Business Impact

- Enables near real-time operational monitoring
- Supports proactive SLA management

- Reduces dependency on delayed manual reporting

Expected Outcome

- Improved operational responsiveness
- Faster escalation handling
- Better SLA compliance monitoring

2.4 Out-of-Scope (Explicitly NOT in Phase 1)

- **ServiceNow Modification:** Dashboard reads data only; does not modify ServiceNow
- **Automated Incident Assignment:** Dashboard monitors; doesn't auto-assign
- **Change Management Dashboard:** Phase 1 is Incident Management only
- **Problem Management:** Tracking problematic categories only; full problem CMDB in Phase 2
- **Mobile App:** Responsive web only; native mobile apps in Phase 2+
- **Predictive Analytics:** Phase 1 is descriptive/diagnostic; prediction in Phase 2
- **Custom ITIL Workflows:** Works with existing ITIL processes; doesn't change them
- **User Authentication:** Phase 1 assumes internal network access; LDAP in Phase 2
- **Advanced Scheduling:** No automated email reports in Phase 1
- **Third-party Integrations:** ServiceNow only in Phase 1

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3. SUCCESS CRITERIA & ACCEPTANCE

Phase 1 (Week 1-2: MVP)

Functional Acceptance Criteria:

- Dashboard displays 4 KPI cards (Total, MTTR, SLA %, Team Score)
- 3 charts display correctly (Trend, Priority, Category)
- Date range filter works (all metrics update when filter changes)
- At least one additional filter works (Priority or Category)
- Export to Excel works (file downloads with data)
- Data loads from sample CSV file without errors
- Dashboard loads in <3 seconds on standard broadband

Data Accuracy Criteria:

- MTTR calculation matches manual verification (sample incidents)
- SLA % matches manual verification
- Incident count matches source data
- No data loss or corruption

User Experience Criteria:

- Filters are intuitive and easy to use
- Charts are readable and not cluttered
- Colors follow industry standards (red=bad, green=good)
- Responsiveness: Filters apply instantly (<500ms)

Documentation Criteria:

- User guide written (how to interpret metrics)
- README updated with all system components
- Code documented with comments
- BRD completed and signed off

Phase 1.5 (Week 3-4: Polish & Integration)

- Live ServiceNow integration (if API access approved)
- All 5 filters working (Team filter added)
- 5 charts displayed (added Team Performance chart)
- Mobile responsive design tested on iPad/tablet
- Export to PDF implemented
- Training materials created

Phase 2 (Week 5-8)

- Automated data refresh from ServiceNow (every 30 min)
- SLA alert system (notify when at-risk)
- Problem Management metrics (MTTF tracking)
- User authentication/role-based access
- Production deployment

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4. BUSINESS METRICS & MEASUREMENT

Primary Success Metrics

Metric 1: Mean Time To Resolution (MTTR)

- Current: 10.2 hours
- Target: ≤7.0 hours
- Measurement: Dashboard shows automatic calculation
- Frequency: Daily visible in dashboard
- Owner: Service Desk Manager

Metric 2: SLA Compliance %

- Current: 82%
- Target: 95%+
- Measurement: Dashboard shows compliance % by priority
- Frequency: Real-time in dashboard
- Owner: IT Director

Metric 3: Help Desk Reporting Time Saved

- Current: 6-8 hours/week
- Target: ≤30 minutes/week
- Measurement: Time tracking; survey
- Frequency: Monthly
- Owner: Service Desk Manager

Metric 4: Dashboard Adoption

- Target: 80%+ of IT staff using within 1 month
- Measurement: Login analytics; usage statistics
- Frequency: Weekly reports
- Owner: IT Director

Metric 5: Incident Escalation Response Time

- Current: Hours (via weekly report)
- Target: <15 minutes (real-time alert)
- Measurement: Time from SLA at-risk alert to manager action
- Frequency: Per incident (Phase 2 with alerts)
- Owner: Service Desk Manager

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5. CONSTRAINTS & ASSUMPTIONS

Constraints

C1: Budget Constraint

- Constraint: $46K total Year 1 cost
- Impact: Scope must fit budget
- Decision: Build with open-source tools, leverage existing infrastructure

C2: Timeline Constraint

- Constraint: Must launch by end of April 2024
- Impact: Limited time for extensive customization
- Decision: MVP first approach, Phase 2 for enhancements

C3: Infrastructure Constraint

- Constraint: Must use existing IT infrastructure (no new servers)
- Impact: Dashboard runs on development machine or shared application server
- Decision: Architecture supports cloud migration in Phase 2

C4: ServiceNow API Access

- Constraint: Requires IT Security approval for API access
- Impact: May start with sample data; migrate to live data later
- Decision: Build with abstraction layer for data source (CSV → API → Database)

C5: No Workflow Changes

- Constraint: Dashboard cannot require changes to incident handling process
- Impact: Dashboard is read-only observer; doesn't change ServiceNow
- Decision: Dashboard reports on existing processes only

Assumptions

A1: Data Quality

- Assumption: ServiceNow incident data is reasonably clean
- Impact: If data quality poor, metrics unreliable
- Mitigation: Data validation rules; flag suspicious records

A2: Stakeholder Availability

- Assumption: IT Manager and Director available for 2-3 requirements meetings
- Impact: Requirements locked after validation
- Mitigation: Comprehensive BRD upfront

A3: Infrastructure Available

- Assumption: Server/hosting available for deployment by Week 5
- Impact: Cannot deploy to production
- Mitigation: Coordinate with IT Infrastructure early

A4: Incident Definition Consistency

- Assumption: "Incident" definition consistent in ServiceNow
- Impact: If definitions vary, metrics skewed
- Mitigation: Document incident definition; monitor for changes

A5: SLA Definitions Stable

- Assumption: SLA times (Critical=2h, High=8h, etc.) won't change mid-project
- Impact: Target metrics invalid if SLA changes
- Mitigation: Document SLA definitions; version them

6. RISKS & MITIGATION

Risk	Probability	Impact	Mitigation
Poor ServiceNow API access	Medium	High	Get pre-approval; design data layer for CSV fallback
Data quality issues	Medium	Medium	Build validation; flag errors; educate on data entry
User adoption low	Low	High	Demonstrate quick wins; show time savings; provide training
Performance issues	Low	High	Load testing; caching strategy; CDN for static content
Schedule slips	Medium	High	Aggressive MVP approach; cut Phase 2 features if needed
Integration challenges	Medium	Medium	Start integration early; allocate time for debugging

7. APPROVAL & SIGN-OFF

Document Approval

Role	Name	Signature	Date
BA / Author	Harshal Bhavsar	___________________	
Service Desk Manager	Sarah Chen	___________________	__________
IT Director	Akshaya Parhi	___________________	__________
Finance	John Williams	___________________	__________
Sponsor	David Brown	___________________	__________

Acknowledgment

By signing above, stakeholders confirm:

- ✓ They have reviewed and approve these requirements
- ✓ They understand what is in-scope and out-of-scope
- ✓ They accept the timeline and resource plan
- ✓ They will support the project during implementation
- ✓ They will participate in user testing and feedback

APPENDIX A: SAMPLE INCIDENT DATA & CALCULATIONS

Sample Data (from 200 incidents in Jan-Apr 2024)

MTTR Calculation Example:

Sample incidents:

- INC001: Created 01-01 08:00, Resolved 01-01 10:30 = 2.5 hours (Critical)
- INC004: Created 01-02 14:30, Resolved 01-02 16:00 = 1.5 hours (Critical)
- INC008: Created 01-04 15:00, Resolved 01-04 16:30 = 1.5 hours (Critical)

MTTR for Critical = (2.5 + 1.5 + 1.5) / 3 = 1.83 hours ✓ TARGET MET

High priority sample:

- INC002: 28.75 hours
- INC006: 4 hours
- INC009: 4 hours

MTTR for High = (28.75 + 4 + 4) / 3 = 12.25 hours (above 8h target) ⚠️

Action: Investigate why INC002 took 28.75 hours

SLA Compliance Calculation Example:

Critical SLA Target: 2 hours

Sample Critical incidents:

- INC001: 2.5 hours → FAILED (exceeds 2h)
- INC004: 1.5 hours → PASSED
- INC008: 1.5 hours → PASSED
- INC011: 2.25 hours → FAILED
- INC017: 1.5 hours → PASSED
- INC021: 0.5 hours → PASSED
- INC027: 1.75 hours → PASSED
- INC031: 2 hours → PASSED
- INC037: 1.5 hours → PASSED
- INC041: 2.5 hours → FAILED

SLA Compliance = 7 compliant / 10 total = 70% ❌ BELOW TARGET

Insight: Critical incidents missing SLA by 10-30 min

Action: Investigate why escalation is slow; improve prioritization

Incident Distribution:

From 200 incidents in sample:

Priority breakdown:

- Critical: 25 (12.5%) - Should be rare; INVESTIGATE IF INCREASING

- High: 50 (25%) - Expected 20-30%

- Medium: 75 (37.5%) - Expected 40-50%

- Low: 50 (25%) - Expected 20-30%

Observation: Critical is 12.5% (higher than expected)

Action: Review categorization; ensure real critical incidents only

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APPENDIX B: ITIL ALIGNMENT

ITIL Processes Supported by Dashboard

Incident Management

- Metrics: MTTR, SLA %, incident volume by priority/category/team
- Support: Real-time visibility enables faster escalation and resolution
- ServiceNow tables: incident (primary)

Problem Management

- Metrics: MTTF (mean time to failure), recurring incident patterns
- Support: Identify problematic categories; track if solutions stick
- Phase 2: Problem ticket creation with context

Service Level Management

- Metrics: SLA compliance %, SLA trend, at-risk incidents
- Support: Real-time SLA tracking; proactive alerts for violations
- ServiceNow tables: service_level, service_level_agreement

Availability Management

- Metrics: Incident impact, availability %, MTBF
- Support: Understand which services have availability issues
- Phase 2: Service availability dashboard

Capacity Management

- Metrics: Team workload, incident volume trend, MTTR by team

- Support: Plan resources based on actual data
- Phase 2: Capacity planning dashboard

Change Management

- Metrics: Incidents post-change, change success measurement
- Support: See if changes cause incidents; measure change quality
- Phase 2: Change impact analysis

APPENDIX C: GLOSSARY OF TERMS

MTTR (Mean Time to Resolution)

Definition: Average time from incident creation to resolution
Formula: Sum of (resolution_time) / Count of resolved incidents
Units: Hours
Importance: Lower is better; directly impacts customer satisfaction
Target: 7.0 hours

SLA (Service Level Agreement)

Definition: Promised response/resolution times by incident priority
SLA Targets at TechServe:

- Critical: 2 hours (service down, >100 users)
- High: 8 hours (major degradation, 10-100 users)
- Medium: 24 hours (minor issue, <10 users)
- Low: 72 hours (single user, workaround available)

SLA Compliance % Definition: % of incidents resolved within their SLA time
Formula: (Incidents resolved within SLA / Total resolved) × 100%
Target: 95%+
Importance: Proves commitment to customers; drives reputation

MTTF (Mean Time To Failure) Definition: Average time between incident resolution and recurrence
Importance: Higher is better; shows if solutions are permanent
Phase 2 metric

Incident Priority

- **Critical:** Service down, large impact (>100 users), requires immediate response
- **High:** Significant degradation, 10-100 users affected, urgent response
- **Medium:** Minor issue, <10 users, normal business hours response
- **Low:** Single user, workaround available, can be deferred

Incident Category

- **Network:** Connectivity, VPN, DNS, routing, WAN issues
- **Access:** Login, permissions, account, authentication issues

- **Hardware:** Physical equipment failure, printers, monitors, laptops
- **Software:** Application, system software, data issues
- **Other:** Miscellaneous, doesn't fit categories above

Incident Status

- **Open:** Created but not yet resolved
- **Resolved:** Fixed but waiting for user confirmation
- **Closed:** User confirmed resolution; no longer active
- **On-Hold:** Waiting for customer/external resource

APPENDIX D: CURRENT STATE STATISTICS

January-April 2024 Incident Summary:

Total Incidents: 200

Average incidents/month: 50

Peak day: Monday mornings (most incidents)

Low day: Weekends (fewer reports)

Priority Distribution:

- Critical: 25 (12.5%) - Should be <5%, investigate

- High: 50 (25%) - Good, expected range

- Medium: 75 (37.5%) - Good, expected range

- Low: 50 (25%) - Good, expected range

Category Distribution:

- Network: 50 (25%) - Largest category

- Hardware: 60 (30%) - Second largest

- Access: 40 (20%)

- Software: 50 (25%)

MTTR by Priority:

- Critical: 1.83 hours (Target: <2h) ✓ MEETING

- High: 5.12 hours (Target: <8h) ✓ MEETING

- Medium: 8.25 hours (Target: <24h) ✓ MEETING

- Low: 28.5 hours (Target: <72h) ✓ MEETING

- Overall: 10.2 hours (Target: <7h) ❌ NEEDS IMPROVEMENT

SLA Compliance by Priority:

- Critical: 88% (Target: 95%) ⚠️ At Risk

- High: 84% (Target: 95%) ⚠️ At Risk

- Medium: 81% (Target: 95%) ❌ Failing

- Low: 76% (Target: 95%) ❌ Failing

- Overall: 82% (Target: 95%) ❌ NEEDS IMPROVEMENT

APPENDIX E: TECHNOLOGY STACK

Frontend: HTML5, CSS3, JavaScript, Plotly.js charts **Backend:** Python 3.9+, Flask web framework **Database:** SQLite (dev), PostgreSQL (production) **API Source:** ServiceNow REST API **Deployment:** Docker containers, Kubernetes orchestration (Phase 2) **Hosting:** Internal server (Phase 1), Cloud (AWS/Azure Phase 2)

END OF DOCUMENT

NEXT STEPS

1. **Review & Approval:** Circulate BRD to stakeholders for review (1 week)
2. **Refinement:** Address any feedback; finalize requirements (1 week)
3. **Kick-off Meeting:** Review with IT Director and Service Desk Manager
4. **Development Start:** Week of April 15, 2024
5. **MVP Review:** Week of April 22, 2024
6. **Launch:** Week of April 29, 2024

Questions? Contact Harshal Bhavsar for clarification on any requirements.